Shubham Raj

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Professional Summary

Data Engineer with **3+ years** of experience designing and **modernizing enterprise-scale** cloud data platforms on **Google Cloud Platform (GCP)**. Experienced in delivering large-scale data migrations, building scalable **ETL/ELT pipelines**, and developing cloud-native data architectures using Python, **SQL**, **PySpark**, **Apache Airflow**, **dbt**, and **Data Vault**. Proven track record of optimizing data processing, enabling enterprise analytics, and delivering reliable, high-quality data solutions across healthcare, insurance, finance, and data center domains.

Experience

Data Engineer | Quantiphi Analytics Solutions Pvt Ltd. *Jan 2023 – Present*

- **Enterprise Data Platform Modernization | Data Center Domain | GCP, dbt & Data Vault**
  - Successfully migrated **1,300+ enterprise tables**, developed **200+ business views**, and delivered **400+ Tableau reports** by integrating data from Oracle, Exasol, DB2, and other enterprise sources into BigQuery, meeting delivery timelines.
  - **Led a team of data engineers** by planning deliverables, reviewing code, resolving technical blockers, and coordinating parallel development across multiple data models.
  - Designed and implemented **Data Vault** (Raw, Business, Gold) models using **dbt**, enabling scalable, maintainable, and analytics-ready data architecture.
  - Developed complex SQL views for **enterprise revenue reporting** by implementing multi-stage validation and reconciliation logic, ensuring **100% data accuracy** and consistent financial reporting.
- **Legacy Data Modernization | Insurance Domain | BigQuery & PySpark**
  - Modernized legacy insurance platforms by converting Mainframe (COBOL) programs into optimized BigQuery SQL, migrating **billions of records** across **30+ enterprise processes** with **100% reconciliation accuracy**.
  - Built scalable **PySpark** transformation frameworks on **Dataproc Serverless** for distributed processing of high-volume datasets, reducing manual intervention.
  - Optimized Spark-based data processing workloads through partitioning strategies, query tuning, and efficient data transformations, reducing execution costs and improving performance by **25%**.
- **Healthcare Data Platform Modernization | Healthcare Domain | BigQuery & Apache Airflow**
  - Built and operated ETL pipelines using **Apache Airflow** (Cloud Composer) to migrate data from **Teradata** to **BigQuery**.
  - Optimized complex SQL transformations, reducing execution time from **12+ hours** to **under 10 minutes** through query tuning and execution plan optimization.
  - Implemented a centralized Airflow monitoring and alerting framework for **150–200 DAGs** across **10–15 business teams**, automating failure notifications and improving SLA compliance.
- **Cloud Data Migration | Finance Domain | Snowflake to BigQuery**
  - Built and deployed end-to-end data migration pipelines using **Informatica IICS** to migrate enterprise datasets from **Snowflake** to **BigQuery**.
  - Validated migrated datasets through **SQL reconciliation**, query optimization, and functional testing across Development and QA environments, ensuring **production readiness**.
  - Developed Tableau dashboards and business reports to support post-migration analytics and stakeholder reporting.

Technical Skills

Programming: Python, SQL, PySpark, COBOL (Legacy), LUA
Cloud & Big Data: Google Cloud Platform (BigQuery, Cloud Storage, Cloud Composer, Cloud Run, Dataproc Serverless)
Data Engineering: ETL/ELT, Data Pipelines, Data Migration, Data Modeling, Data Vault, dbt, Apache Spark, Data Validation / Quality
DevOps & BI Tools: Apache Airflow, CI/CD, Git, Github Actions, Bitbucket, Tableau
Databases & Data Warehouses: BigQuery, Snowflake, Teradata, Exasol, DB2, MySQL

Certifications

Professional Data Engineer | Google Cloud Certified *Dec 2024*
Associate Cloud Engineer | Google Cloud Certified *May 2023*
Oracle Cloud - Foundations Associate | Oracle Certified *Nov 2023*
Oracle Cloud - AI Foundations Associate | Oracle Certified *Nov 2023*

Education

National Institute of Technology, Durgapur *Jul 2019 – Jun 2023*

- B.Tech - Metallurgical and Materials Engineering